COCA-COLA | SENSORY & BLIND TASTE INTELLIGENCE PLATFORM

Openv1_blind_test Process Flow, Connections & Runbook

Webs

Live endpoints

Local dev : http://localhost:5174/

Production (Vercel) : https://coca-sensory.vercel.app

Mirror : https://coca-sensory-3yh4wxwil-kushik12debnath-bits-projects.vercel.app

GitHub : https://github.com/kushik12debnath-bit/Openv1_blind_test

Desktop copy : C:\Users\Kaushik\Desktop\Openv1_blind test\

Source repo : C:\Users\Kaushik\coca-sensory\

Demo credentials (role-wise)

Username	Password	Role + access
admin	admin123	Admin: Terminal + Reveal + Dashboard + CSV/PDF
executive	exec123	Executive: Reveal + Dashboard + exports
tester	test123	Tester: Terminal + Reveal only

Auth is demo-grade client-side only (src/lib/auth.ts, sessionStorage). Invalid logins are rejected with an inline error. This document describes every screen flow, every connection between code, data, and deployments, and the exact verification performed.

1. System map (what connects to what)

1.1 One-picture architecture

Browser (React SPA, Vite) --> In-memory state (evaluations[1240 + new], filters, session)
|--> Charts render from filtered state (Recharts, no backend calls)
|--> CSV export serializes filtered state to a downloaded file
|--> PDF report opens a print sheet from filtered state (window.print)
|--> Auth gate reads src/lib/auth.ts role table + sessionStorage
Build (tsc + vite) --> dist/ --> Vercel CDN (production URL)
Source (C:\Users\Kaushik\coca-sensory) --> git --> GitHub Openv1_blind_test --> Vercel project coca-sensory
Source --> robocopy --> Desktop copy Openv1_blind test (source-only, needs npm install)

1.2 Ports & environments

Env	URL	Notes
Local Vite	http://localhost:5174/	npm run dev -- --port 5174
Prod Vercel	https://coca-sensory.vercel.app	READY, HTTP 200 verified
Prod mirror	https://coca-sensory-...vercel.app	Direct deployment URL
GitHub	github.com/.../Openv1_blind_test	Public, branch main

1.3 Code map (file -> responsibility)

- src/App.tsx: session restore, role gating, module switch, toast bus, CSV/PDF handlers.
- src/lib/auth.ts: DEMO_USERS, authenticate(), canAccess(), allowedModules(), canExport(), session save/restore.
- src/components/Login.tsx: login form, quick-fill role buttons, invalid-credential error.
- src/components/chrome.tsx: TopNav (role-filtered nav + user badge + logout), KpiCard, FilterBar, GlassTooltip, ToastStack.
- src/components/Terminal.tsx: Module A blind terminal (batch codes, 6 sliders x2, 15s rinse, forced choice, metadata).
- src/components/Reveal.tsx: Module B spring flip (stiffness 260, damping 20), Coke vs Competitor cards, baseline bars.
- src/components/Dashboard.tsx: Module C KPIs + Donut/Radar/Bar/Area (ResponsiveContainer 100%x320) + export buttons.
- src/components/ReportModal.tsx: executive print sheet (#executive-print-sheet) with KPIs, chi-square table, snapshots.
- src/lib/sensory.ts: types, seed dataset N=1240 (68.4% Coke), filters, summaries, baselines 8.9 vs 7.1.
- src/lib/export.ts: RFC-4180 CSV builder + browser download, print trigger.
- vercel.json / vite.config.ts / tailwind.config.js: SPA rewrites, build, brand tokens (#F40009, #09090B, #F8FAFC).

2. Auth & role-wise access flow

2.1 Login sequence

1) App boots -> restoreSession() checks sessionStorage key coca-sensory-session-v1.
2) No session -> <LoginScreen/> (branded split panel, username + password, quick-fill buttons).
3) Submit -> authenticate(username, password) exact-matches DEMO_USERS (case-insensitive username).
4) Fail -> inline error: 'Invalid credentials. Use one of the demo role accounts below.' Session stays logged-out.
5) Success -> saveSession(), allowedModules(role) computed, landing = terminal if allowed else first allowed module, toast 'Signed in as X'. TopNav shows 'username - ROLE' + Log out.

2.2 Permission matrix (enforced in UI)

Capability	admin	executive	tester
Terminal submit	YES	NO (hidden)	YES
Reveal view	YES	YES	YES
Dashboard BI	YES	YES	NO (hidden)
CSV export	YES	YES	BLOCKED+toast
PDF report	YES	YES	BLOCKED+toast

Nav is filtered: NAV_ITEMS.filter(item => allowed.includes(item.id)). Direct module changes go through handleModuleChange() which re-checks canAccess() and toasts 'Access restricted' on violation. Dashboard/Terminal bodies double-check canAccess before rendering. Tester banner shows 'Exports disabled for tester role'. Logout clears sessionStorage, lastDraft, and report modal.

2.3 Verified login test log (browser automation, 2026-10-16)

Test	Input	Observed	Result
Admin login	admin/admin123	ADMIN, modules terminal-reveal-dashboard, N=1,240	PASS
Executive login	executive/exec123	EXECUTIVE, modules reveal-dashboard, no terminal	PASS
Tester login	tester/test123	TESTER, modules terminal-reveal, exports disabled	PASS
Invalid login	hacker/wrong	Inline invalid-credentials error, stays on login	PASS
Logout	Log out button	Returns to Executive demo default login	PASS

3. Module flows (A / B / C)

Module A - Live Blind Evaluation Terminal (tester + admin)

Entry: TopNav 'Blind Evaluation'. Two masked samples Batch #AAA and #BBB (random 100-999, distinct; secret cokeIsA flag decides which hides Coca-Cola). Each sample card has six 1-10 sliders: Carbonation Bite, Brix/Sweetness, Acidity/Citrus, Caramel/Spice Aroma, Mouthfeel/Viscosity, Aftertaste Cleanliness, with live badge + card mean. Center Palate Cleanser card runs a 15s countdown (Start/Restart rinse, progress bar, Ready state) encouraging a water rinse between samples. Forced-choice offers Batch #A / Batch #B / Tie; metadata offers Age (18-24, 25-34, 35-44, 45+), Frequency (Daily/Weekly/Occasional), Location (Mall Kiosk, University Campus, Corporate Lounge). Submit is disabled until a choice is picked. On submit the secret mapping converts display pick to BrandChoice (coca-cola/competitor/tie), computes cokeMean/compMean, prepends the Evaluation (id SES-xxxxxx, ISO timestamp), opens Reveal, fires toast, and mints a fresh batch pair.

Module B - Unblinding & Brand Reveal (all roles)

Entry: TopNav 'Unblinding Reveal' (pulsing dot when a fresh ballot is pending). Empty state ('No sealed ballot yet' + CTA) shows if nothing was submitted this session. Otherwise a dark header shows ballot id, timestamp, location, age, frequency. Two cards flip in with Framer Motion spring (type spring, stiffness 260, damping 20, second card delayed 0.12s, perspective 1200): left red-gradient Coca-Cola Original Taste card (official SVG logo on white, secret #cokeCode, your coke mean, per-metric you-vs-baseline bars) and right slate-blue Competitor card (secret #compCode, your comp mean, same bars). Verdict strip compares your gap to the national gap (8.9 vs 7.1 = 1.8 pts): tie message, aligned message (68.4% also chose Coke), or contrarian message, plus CTA into telemetry.

Module C - Executive Telemetry Dashboard (admin + executive)

Entry: TopNav 'Executive Telemetry'. FilterBar (Location / Age / 24H-7D-30D-All) drives applyFilters() -> summarize() for every KPI and chart. KPI ribbon: Net Preference Share (cokeShare %, +14.2% vs Q3 pill), Statistical Significance (p<0.001, 99.9% CI chi-square), Mean Organoleptic Index (cokeMean vs compMean, delta), Blind-to-Brand Conversion (loyalty %). Charts (each in fixed 320px parent + ResponsiveContainer 100%x320): Donut (Coke #F40009 / Competitor #3B82F6 / Tie #94A3B8, center cokeShare% + N), Radar (6 axes, Coke vs Competitor means), Bar (per-cohort Coke/Competitor/Tie %), Area spline (10-day cumulative counts per location). Empty-state guards replace every chart when filters match zero rows. Export row offers CSV telemetry + Executive PDF report buttons (export-gated by role).

4. Data, exports & report engine

4.1 Seed dataset (src/lib/sensory.ts)

generateSeedEvaluations(1240) uses a mulberry32 PRNG (seed 20260214) so the demo is stable: 68.4% coca-cola / 24.6% competitor / ~7% tie; locations 38/33/29; ages 28/34/22/16; frequencies 42/38/20; timestamps spread over 14 days, sorted newest-first. Per-metric scores jitter around NATIONAL_BASELINE_COKES [8.8,9.0,8.7,8.9,8.8,9.1] and NATIONAL_BASELINE_COMP [7.0,7.3,6.9,7.2,7.0,7.2] with winner-dependent offsets, clamped 1-10. New terminal submissions prepend live and instantly move every KPI/chart because they derive from the same evaluations array.

4.2 Evaluation record

id, timestamp (ISO), location, ageCohort, frequency, batchCoke (secret Coke code), batchComp (secret competitor code), cokeScores[6], compScores[6], preference (coca-cola|competitor|tie), cokeMean, compMean.

4.3 CSV export (evaluationsToCsv, RFC-4180)

Header: session_id, timestamp, location, age_cohort, consumption_frequency, batch_coke_code, batch_competitor_code, coke_carbonation..coke_aftertaste (6), competitor_* (6), coke_mean, competitor_mean, winning_brand. Values quoted per RFC-4180, CRLF line endings. Filename: coca-cola-blind-test-telemetry.csv. Exports the currently filtered set; toast confirms row count. Testers are blocked with an 'Export blocked' toast.

4.4 Executive PDF report (ReportModal + print CSS)

Button 'Executive PDF report' (dashboard) or 'Generate Executive PDF' opens #executive-print-sheet modal: logo header, generation timestamp, N, 4 KPI tiles, chi-square summary table (observed vs null-33.3% expected per outcome + verdict line), preference-split bars snapshot, per-metric means snapshot. Print button calls window.print(); @media print CSS isolates the sheet (body hidden, sheet absolute, .no-print controls removed). Operator picks 'Save as PDF', 1 page, background graphics ON. Testers are blocked with a 'Report blocked' toast.

5. Deployments, runbook & verification

5.1 Connection pipeline

- Local source C:\Users\Kaushik\coca-sensory --npm run dev--> http://localhost:5174 (HMR, verified).
- Local source --npm run build (tsc -b + vite, 2913 modules, ~5s)--> dist/ (git-ignored).
- Local source --git init/add/commit--> GitHub kushik12debnath-bit/Openv1_blind_test, branch main (public).
- Local source --robocopy (excl. node_modules/.git/dist)--> Desktop Openv1_blind test (needs npm install to run).
- Local source --vercel --prod --yes--> Vercel project coca-sensory --> https://coca-sensory.vercel.app (READY).
- Vercel build mirrors local: npm install (148 pkgs) + npm run build on 2-core/8GB iad1 runner.
- Git auto-link showed 'Failed to connect Openv1_blind_test' warning; upload deploy succeeded. Link manually in Vercel Dashboard > coca-sensory > Settings > Git for auto-deploys.

5.2 Run locally (either copy)

1) cd into C:\Users\Kaushik\coca-sensory OR C:\Users\Kaushik\Desktop\Openv1_blind test
2) npm install
3) npm run dev -- --port 5174 (open http://localhost:5174)
4) Sign in: admin/admin123 (full), executive/exec123 (BI), tester/test123 (terminal).
5) Terminal -> pick a batch -> Submit -> Reveal flip -> Dashboard telemetry moves. Dashboard -> CSV / PDF buttons.

5.3 Verification log (this session)

Check	Command / action	Evidence	OK?
Build	npm run build	tsc+vite, 2913 modules, 4.98s, 0 TS errors	YES
Local HTTP	curl localhost:5174	status:200, correct title	YES
Prod HTTP	curl coca-sensory.vercel.app	status:200	YES
GitHub push	gh repo create+push	Openv1_blind_test, HEAD->main	YES
Desktop copy	robocopy+rename	Desktop\Openv1_blind test, 31 files	YES

5.4 Limits & next steps

- Auth is demo-only (plaintext demo passwords in src/lib/auth.ts). For production add a real backend (OAuth/OIDC), hashed passwords, and server-side role checks.
- Telemetry lives in browser memory; refresh resets to the 1240 seed. Persist via API + database for multi-user panels.
- Bundle is ~850KB JS (Recharts + Framer). Code-split Dashboard/Terminal/Reveal with React.lazy for faster first paint if needed.
- To re-deploy: edit -> npm run build -> git add/commit/push -> vercel --prod --yes (or auto-deploy once Git is linked).